

HR Analytics Dashboard

Prepared By

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Abstract

Human Resource Analytics plays a crucial role in helping organizations understand workforce trends and make informed business decisions. This project focuses on developing an interactive HR Analytics Dashboard using Power BI to analyse employee data and provide actionable insights related to workforce demographics, employee satisfaction, promotion planning, retrenchment analysis, and organizational performance.

The dashboard transforms raw employee data into meaningful visualizations and Key Performance Indicators (KPIs), enabling Human Resource departments to make data-driven decisions. Through data transformation, modelling, and visualization techniques, the dashboard provides a comprehensive view of employee distribution, promotion eligibility, workforce stability, and employee engagement.

1. Introduction

Organizations generate large amounts of employee data that can be utilized to improve workforce planning and HR decision-making. Traditional reporting methods often make it difficult to identify workforce trends and monitor employee performance effectively.

This project aims to develop an HR Analytics Dashboard that consolidates employee information into a single interactive reporting solution. The dashboard enables HR managers to monitor employee demographics, identify promotion opportunities, analyse workforce distribution, and track employees approaching retirement or retrenchment.

2. Problem Statement

Human Resource departments require efficient tools to monitor employee performance, workforce demographics, promotion cycles, and retirement planning.

Without proper analytical tools, organizations face challenges such as:

- Difficulty identifying employees due for promotion.
- Lack of visibility into workforce demographics.
- Inefficient workforce planning.
- Limited understanding of employee satisfaction.
- Challenges in monitoring retirement and retrenchment requirements.

The HR Analytics Dashboard addresses these issues through interactive visual reporting and KPI monitoring.

3. Objectives

The main objectives of this project are:

- Analyse employee demographics.
 - Monitor workforce distribution.
 - Identify employees due for promotion.
 - Track employees approaching retirement or retrenchment.
 - Analyse employee satisfaction levels.
 - Evaluate workforce performance indicators.
 - Provide actionable insights for HR decision-making.
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4. Dataset Description

The dataset contains information about employees working within an organization.

Important Attributes

- Employee Number
- Gender
- Age
- Department
- Job Role
- Job Level
- Education
- Monthly Income
- Job Satisfaction
- OverTime
- Total Working Years
- Years At Company
- Years Since Last Promotion
- Distance From Home
- Attrition

Dataset Size

- Total Employees: 1470
- Male Employees: 882
- Female Employees: 588

5. Data Preparation and Transformation

Data preparation was performed using Power Query Editor in Power BI.

Power Query Activities

- Imported HR employee dataset into Power BI.
- Validated data types for all columns.
- Renamed columns for consistency and readability.
- Checked for missing values and duplicate records.
- Removed unnecessary fields.
- Applied filtering and formatting transformations.
- Created custom columns for business analysis.
- Standardized data formats.
- Prepared data for KPI calculations and reporting.

Benefits of Data Transformation

- Improved data quality.
 - Reduced reporting inconsistencies.
 - Enhanced dashboard performance.
 - Enabled accurate business analysis.
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6. Data Modelling

After data transformation, the dataset was loaded into Power BI for modeling and analysis.

Measures and Calculations

Several DAX measures were created to calculate:

- Total Employees
- Male Employees
- Female Employees
- Due for Promotion
- Due for Retrenchment
- Active Employees
- Employee Ratings
- Workforce Percentages

These measures were used to generate dynamic visualizations and KPIs.

7. Dashboard Architecture

The dashboard consists of three interconnected pages:

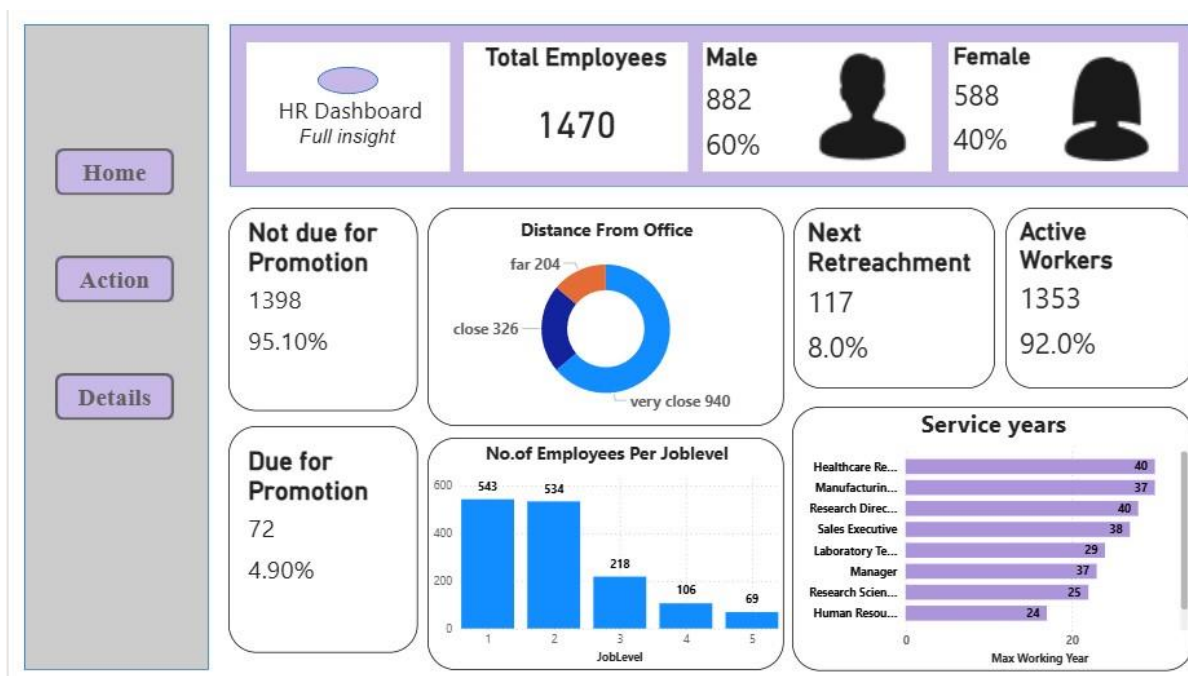
1. Home Dashboard
2. Action Dashboard
3. Details Dashboard

Navigation buttons were implemented to allow users to switch between pages easily.

7.1 Home Dashboard

Purpose

Provide an executive overview of workforce demographics.



Total Employees

Count: 1470 Employees

This KPI card displays the total workforce available in the organization. It serves as the primary metric for workforce analysis and provides the baseline for all HR-related calculations and reporting.

Male Employees

Count: 882 Employees (60%)

This KPI highlights the male workforce population within the organization. The analysis helps HR understand workforce demographics and gender representation.

Female Employees

Count: 588 Employees (40%)

This KPI represents the female workforce population and supports diversity and workforce composition analysis.

Due for Promotion

Count: 72 Employees (4.9%)

This KPI identifies employees who have met the organization's promotion criteria. It helps HR teams monitor career progression and prepare promotion-related decisions.

Not Due for Promotion

Count: 1398 Employees (95.1%)

This KPI represents employees who currently do not qualify for promotion. The high percentage indicates that only a limited number of employees are currently eligible for advancement.

Next Retrenchment

Count: 117 Employees (8%)

This KPI highlights employees approaching retirement or retrenchment criteria. It supports workforce planning and helps HR prepare succession strategies.

Active Workers

Count: 1353 Employees (92%)

This KPI shows the number of employees actively contributing to the organization. The high percentage indicates a stable and productive workforce.

Distance from Office Analysis

Visual: Donut Chart

Values:

- Very Close: 940 Employees
- Close: 326 Employees
- Far: 204 Employees

This chart analyses employee commuting distance from the workplace. Most employees live very close to the office, indicating convenient workplace accessibility and potentially reduced commuting-related challenges.

Employees per Job Level

Visual: Column Chart

Values:

- Level 1: 543 Employees
- Level 2: 534 Employees
- Level 3: 218 Employees
- Level 4: 106 Employees
- Level 5: 69 Employees

This chart illustrates workforce distribution across organizational job levels. The majority of employees belong to Levels 1 and 2, indicating a workforce concentrated at entry and mid-level positions.

Service Years Analysis

Visual: Bar Chart

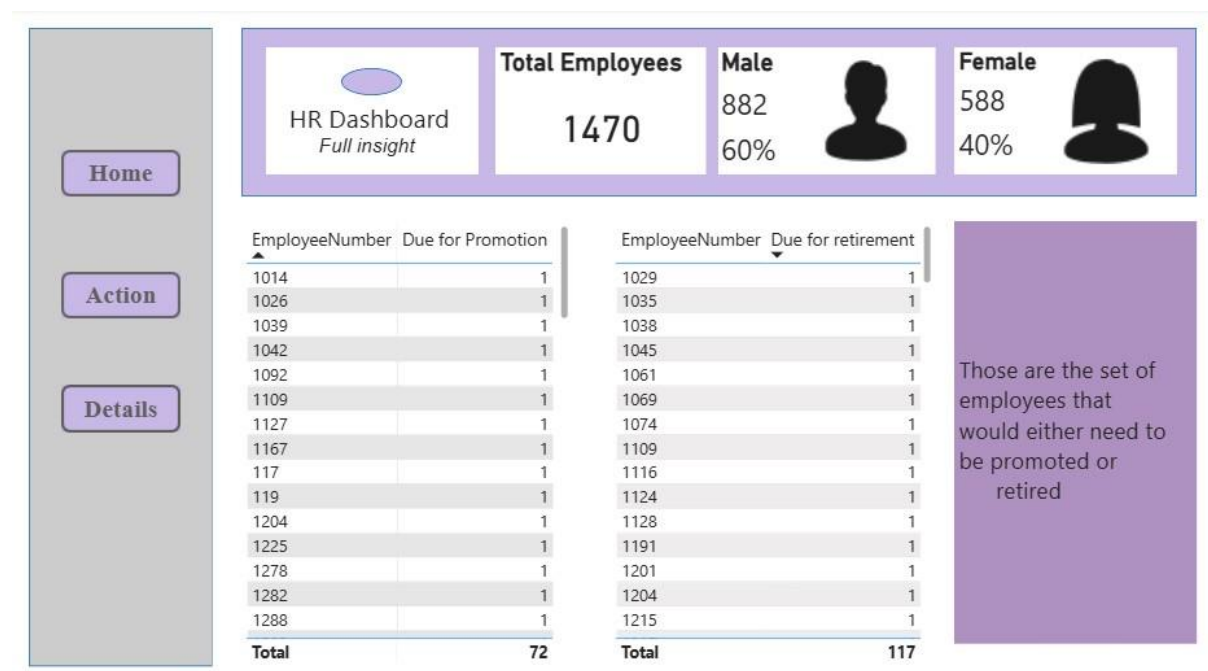
This chart displays the maximum years of service across different job roles.

The analysis reveals that several roles contain highly experienced employees with long tenures, supporting workforce stability and organizational knowledge retention.

7.2 Action Dashboard

Purpose

Support workforce planning and HR decision-making.



Employees Due for Promotion

Count: 72 Employees

This table visual displays employee records identified as eligible for promotion. It enables HR managers to review employee career progression and plan promotion cycles efficiently.

Employees Due for Retirement

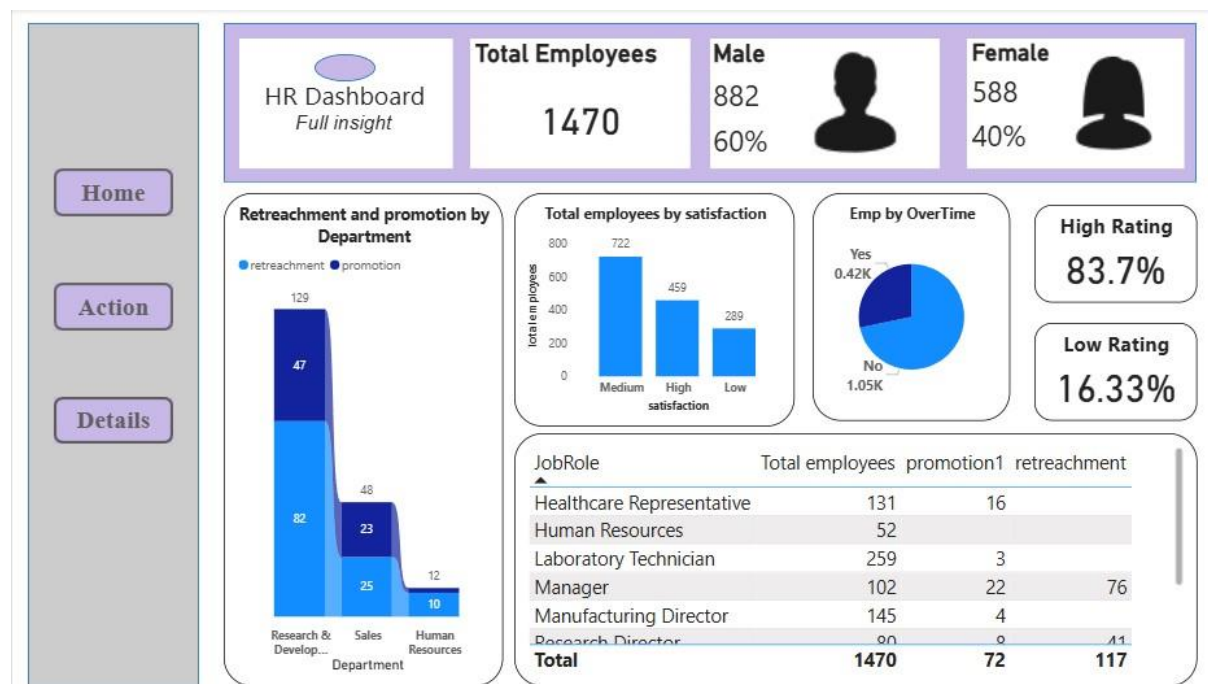
Count: 117 Employees

This table displays employees approaching retirement eligibility. The analysis supports succession planning and workforce replacement strategies.

7.3 Details Dashboard

Purpose

The Details Dashboard provides deeper workforce analytics and performance insights.



Promotion and Retrenchment by Department

Visual: Stacked Column Chart

Values:

Research & Development

- Promotion: 47 Employees
- Retrenchment: 82 Employees

Sales

- Promotion: 23 Employees
- Retrenchment: 25 Employees

Human Resources

- Promotion: 2 Employees
- Retrenchment: 10 Employees

This chart compares workforce actions across departments. Research & Development shows the highest number of employees requiring promotion reviews and retirement planning, indicating greater workforce management requirements within the department.

Employee Satisfaction Analysis

Visual: Column Chart

Values:

- Medium Satisfaction: 722 Employees
- High Satisfaction: 459 Employees
- Low Satisfaction: 289 Employees

This analysis evaluates employee satisfaction levels across the organization. Most employees fall within the medium satisfaction category, while a significant number report high satisfaction levels. A smaller group belongs to the low satisfaction category, highlighting potential opportunities for employee engagement improvement.

Employee Overtime Analysis

Visual: Pie Chart

Values:

- No Overtime: 1054 Employees
- Overtime: 416 Employees

This chart analyses overtime participation among employees. The majority of employees do not work overtime, indicating balanced workload distribution across the workforce.

Employee Rating Analysis

High Rating Employees: 83.7%

Low Rating Employees: 16.33%

These KPI indicators evaluate workforce performance. The results show that most employees maintain strong performance ratings, reflecting a productive and efficient workforce.

Job Role Analysis

Visual: Matrix Table

This table provides a role-wise breakdown of:

- Total Employees
- Employees Due for Promotion
- Employees Due for Retrenchment

The analysis enables HR managers to identify job roles requiring workforce actions, promotion planning, and future succession management. It provides detailed visibility into workforce requirements at the role level and supports strategic HR planning.

8. Key Insights

Workforce Demographics

- Total employees: 1470
- Male employees: 60%
- Female employees: 40%

Promotion Planning

- 72 employees are due for promotion.
- 95% of employees are not currently promotion eligible.

Retrenchment Planning

- 117 employees are approaching retirement or retrenchment criteria.

Employee Satisfaction

- Majority of employees have medium or high satisfaction ratings.

Workforce Stability

- 92% of employees remain active workers.

Department Analysis

- Research & Development is the largest department.
 - Human Resources is the smallest department.
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9. Business Impact

The dashboard helps organizations:

- Improve workforce planning.
- Identify promotion opportunities.

- Monitor retirement and retrenchment requirements.
 - Track employee satisfaction.
 - Reduce manual reporting efforts.
 - Enable data-driven HR decision-making.
 - Improve operational efficiency.
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10. Tools and Technologies Used

Power BI

- Dashboard Development
- Data Visualization
- Interactive Reporting

Power Query

- Data Cleaning
- Data Transformation
- Data Preparation

DAX

- KPI Calculations
- Custom Measures
- Business Logic

Microsoft Excel

- Data Storage
 - Data Validation
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11. Conclusion

The HR Analytics Dashboard successfully converts employee data into actionable business insights. By integrating Power Query transformations, DAX calculations, and interactive visualizations, the dashboard enables effective workforce management and strategic HR planning.

The project demonstrates practical implementation of Business Intelligence concepts and highlights how Power BI can be used to support Human Resource decision-making through data-driven analysis.

Future Scope

- Predict employee attrition using Machine Learning.
 - Develop automated workforce forecasting models.
 - Integrate real-time HR databases.
 - Create employee performance prediction dashboards.
 - Implement advanced workforce planning analytics.
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